

Project Title	Group Size	Andrew ID 1	Andrew ID 2	Andrew ID 3	Member 1	Member 2	Member 3	Mentor	Checkin completed
Accelerate Diffusion Transformer via Quantization		3 shuaiweh	feiyang2	jiyouz	Shuaiwei Huang	Feiyang Wu	Jieyou Zhao	Mengdi Wu	
Accelerating Nano vLLM Inference via Speculative Decoding		3 oanaveka	pvenkatr	sidhartb	Om Anavekar	Pranav Venkatra	Sidharth Babu	Haojia Sun	1
Accelerating Video Generation with Variable Rate Attention		3 pohsuanh	sonick	wentaoc	Po-Hsuan Huang	Sonic Kuo	Wentao Cao	Zhuoming Chen	
Adaptive Cascaded Speculative Decoding for Efficient LLM Inference		3 gaokaiz	melodyyi	zichuwu	Gaokai Zhang	Yue Yin	Zichu Wu	Yixin Dong	
Adaptive Dynamic Locality-Aware Fair Scheduling in Multi-Tenant LLM Serving		3 yangh3	emilyho	annz	Katherine Hu	Emily Ho	Quinn Zhu	Mengdi Wu	
Adaptive Expert Placement for Efficient MoE Inference		3 devanshu	iwatts	krishv	Devanshu Shah	Ishaan Watts	Krish Vijayan	Zhuoming Chen	
Adaptive KV Cache Compression with Importance-Aware Precision		2 garysu	xinranl3		Gary Su	Lilian Li		Zhuoming Chen	
Approximation-Tolerant Token Routing for Efficient Reasoning Inference		2 adiberna	dscholz		Alessandro Di Bernardo	Danilo Scholz		Zhuoming Chen	
Asymmetric KV Cache Compression for MEDUSA Tree-Based Speculative Decoding		2 ebrownhi	adfusco		Edward Brownhill	Alessandro Fusco		Siyuan Chen	
Automated Kernel Optimization for Transformers Using TVM		2 alvinw2	davidzhu		Alvin Wang	David Zhu		Mengdi Wu	
Automated Kernel Workload Profiling Pipeline for GPU Kernel Benchmarking		2 zhifengl	luw		Zhifeng Li	Chenlu Wang		Yixin Dong	
Benchmarking Attention Alternatives		3 janelu	junjiehu	kbouakl	Jane Lu	Junjie Huang	Kareem Bouakl	Mingkuan Xu	
Benchmarking Diverse KV Cache Compression Paradigms for Long-Context LLM Serving		3 lichil	yichenla	emilyshe	Lichi Lin	Yi-Chen Lai	Emily Shen	Yixin Dong	
Benchmarking Speculative Decoding on Blackwell GPUs		1 tianqix			Tianqi Xu			Zepeng Zhao	
Block-Based KV Cache Management for LLM Serving		2 jiachez	lilyli		Jiache Zhang	Li Li		Haojia Sun	1
Building Privacy-Aware ML Systems: Federated Learning, Differential Privacy, and Secure Aggrega		1 ellal			Ella Li			Mingkuan Xu	
Characterizing Performance Boundaries of Speculative Decoding: When Does It Help and When C		3 anaghap	shreyamu	trasul	Anagha Pai	Shreya Mukherje	Tanvir Rasul	Zepeng Zhao	
Chasing Control-Plane Slack in Persistent GPU Megakernels		1 nikhilag			Nikhil Agarwal			Zepeng Zhao	
ClusterKV: Query-Adaptive Clustered KV Cache Compression Pipeline for Long-Context LLM Infe		3 erubinst	xinyuh2	yzuo2	Esme Rubinstein	Xinyu Hu	Yingjie Zuo	Yixin Dong	
Combining KV Cache Sparsification and Quantization for Self-Speculative Decoding		3 yuehanh	zhanminl	skhatavk	Harry Hu	Jerry Liang	Soham Khatavka	Mingkuan Xu	
Comparing Block Floating Point and Online Codebook Quantization for LLM KV-Cache Compressi		3 aanyak	dhrupv	tomgao	Aanya Kochhar	Dhruv Patel	Tom Gao	Mingkuan Xu	
Content-Aware Paged KV Cache Management for Reasoning Vision-Language Models		2 ksnair	sdharod		Kartik Nair	Sahil Dharod		Haojia Sun	
Current Progress in LLM Reinforcement Learning Systems and Deployment-Focused Optimizatio		2 furix	johnzhan		Furix	John Zhang		Zikun Li	
Decode-Aware Activated-Expert Balancing for vLLM MoE Serving		1 ranitg			Ranit Gupta			Zepeng Zhao	
Designing and Evaluating Joint Inference and Autoscaling Policies for Efficient LLM Serving on Ku		1 terrytu			Terry Tu			Siyuan Chen	
Deterministic GEMM via Latency-Aware Fixed-Order Split-K		3 hongyil2	xinyuj2	zhiaoj	Hongyi Liu	Xinyu Jiang	Zhiyao Jiang	Zepeng Zhao	
Directed Acyclic Graph Optimization		3 yichingc	yosenl	chihninc	Yi-Ching Chen	Yosen Lin	Andrew Chen	Mengdi Wu	
Does Speculator Choice Matter at the Action Level? A Systematic Study of Speculator Selection fo		2 jeannieo	yuzhaot		Jeannie Ou	Yuzhao Tu		Siyuan Chen	
Dynamic Hidden State Heap Folding for Efficient Attention (HFold)		3 adewan2	pram	knody	Aditya Dewan	Pranav Ram	Krish Mody	Zikun Li	
Dynamic Multi-LoRA Inference Speedup via Popularity-Based Merging in vLLM		3 yunany	yunzhu	siyuanma	Allen Yang	Yun Zhu	Siena Ma	Zepeng Zhao	
Efficient Inference Engine for Interactive World Models		1 bnullamo			Bhuvan Nallamothu			Yixin Dong	
Efficient JVP-Enabled Attention Kernels for Improved Flow Map Training		3 cwan2	sheels	ssaxena2	Chenxi Wan	Sheel Shah	Shashwat Saxen	Yixin Dong	
Efficient KV Cache Compression for LLM Serving		1 arjunw			Arjun Walla			Mengdi Wu	
Efficient LLM Inference via the Synergy of Knowledge Distillation and Quantization		1 siyuandu			Siyuan Du			Siyuan Chen	
Efficient Systems for Simulation-to-Real Closed-Loop Learning		3 jhurewit	mrarmadas	muxinl	Jack Hurewitz	Manigandan Ran	Ray Liu	Siyuan Chen	
Efficient Work Scheduling		3 achopra2	jonghyuc	saranshm	Aditya Chopra	Jonghyun Choe	Saransh Malik	Siyuan Chen	
Elastic Resource Allocation for Multi-LLM Serving in Dynamic Cloud Environments		2 yixuanm	zixuanc3		Yixuan Mei	Zixuan Chen		Mengdi Wu	
Energy-Aware Evaluation of KV Cache Compression for Long-Context LLM Inference		3 mpindur	ovashish	yinglinh	Mark Pindur	Ojas Vashishtha	Yung-Lin Hsieh	Zikun Li	
Evaluating Speculative Decoding Methods for Efficient LLM Inference		3 ehtang	yunchuc	yuyia	Eric Tang	Yunchu Chen	Yuyi Ao	Haojia Sun	1
Extending FlashInfer-Bench to New LLM Architectures		3 saigopak	sdhavala	sihanhe	Sai Gopal Reddy Kovvu	Shashank Dhava	Sihan He	Yixin Dong	
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From LLM Serving to Agentic Serving: I/O-Aware KV Cache Scheduling for Tool-Calling Workloads		1 jier			Jie Ren			Haojia Sun	
Generalized Learned Cost Model for CPU/GPU Heterogeneous LLM Inference Optimization		3 xlii8	xinhaota	zhikaih	Xiang Li	Xinhao Tan	Zhikai Hu	Mingkuan Xu	
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Google Graph Scheduling Competition Track A: Systems Engineering		3 hong	tylery	ukc	Haolin Cong	Tyler Yang	Unmesh Chakra	Mengdi Wu	
Hardware-Aware Model Compression Profiling		3 mteague	jscriffi	gdiwan	Madison Teague	Jacob Scriffiny	Gautam Diwan	Siyuan Chen	
How Reliable Are LLM-Generated CUDA Kernels? A Multi-Layer Verification Study		1 demiw			Demi Wang			Yixin Dong	
Hybrid KV-Cache Compression for Efficient Long-Context LLM Inference		3 ageorgev	changxig	dkapur	Andrew Georgev	Changxiang Gao	Dhruv Kapur	Haojia Sun	
Hybrid Predictive KV Cache Eviction for Long-Context LLMs		3 prabhatp	gkesavas	hongwens	Prabhat Pandey	Guhan Kesavasa	Hong-Wen Shen	Zhuoming Chen	

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Investigating System Optimizations in Modern 3D Perception	3	polinc	ctuan	mehon	Po-Lin Chen	Chaol Tuan	Manyung Emma	Yixin Dong		
KV Cache Compression for Model Serving	3	kadinz	ryanlau	stevenl	Kadin Zhang	Ryan Lau	Steven Lee	Zikun Li		
Layer and Semantic-Aware KV Cache Quantization	3	jingfenc	yibingwa	yuhangze	Jingfeng Chen	Yibing Wan	Yuhang Zeng	Zikun Li		
Lightweight Scheduling and Kernel Optimizations for Heterogeneous LLM Serving	3	akshana	acarmi	ashmanm	Akshan Agrawal	Aryeh Carmi	Ashman Mehra	Mingkuan Xu		
Memetic Reflective Evolution for LLM-Driven DAG Scheduling in Deep Learning Compilers	1	went			Wen Tian			Mingkuan Xu		
Memory-Aware Graph Scheduling for ML Compilers	3	andrewz2	houkinw	zhixinc	Andrew Zhang	Hou Kin Wan	Zhixin Chen	Mengdi Wu		
Memory-Efficient Long-Context LLM Inference via KV Cache Compression	3	yanyanw	yujingw	yuningt	Yanyan Wang	Yujing Wang	Yuning Tao	Haojia Sun	1	
Model Compression as a Systems Design Choice for Efficient Inference	3	haotongx	wenhanli	xinyay	Haotong Xin	Wenhan Li	Xinya Yan	Haojia Sun	1	
Optimization and Performance Analysis of Keyword Spotting (KWS) on RISC-V Platforms	3	shengfeg	kangjie1	yantingj	Shengfei Gu	Kangjie Lu	Yanting Ji	Shushi Hong		
Optimizing FP8 Fused Mixture-of-Experts Kernels for LLM Inference on NVIDIA Blackwell GPUs	3	chengyec	chinjuic	ethanchi	Cheng-Yeh Chung	Chin Jui Chen	Ethan Chiu	Zepeng Zhao		
Optimizing LLM Inference Using Context Distillation	1	seungjio			John Oh			Zikun Li		
Optimizing LLM Serving with Speculative Decoding	3	cgao3	yuchenz8	zihaidu	Chenhao Gao	Yuchen Zhou	Zihao Du	Siyuan Chen		
Optimizing MXFP4 Mixture-of-Experts Kernels on AMD MI300X	1	diac1			Diac Liu			Zepeng Zhao		
Optimizing Sparse Attention Kernels for Long-Context Inference	1	mingjuns			Mingjun Shen			Shushi Hong		
Ordering Effects of KV Cache Compression under Grouped-Query Attention	3	ayushib	miaoyany	sophiezh	Ayushi Bansal	Miaoyang Ye	Sophie Zhao	Shushi Hong		
OT-KV: Information-Preserving KV Cache Compression via Optimal Transport with Bias-Corrected	3	yini	xinkunw	zzheng4	Yi Ni	Xinkun Wang	Zhiheng Zheng	Shushi Hong		
Paged-TierKV: Block-Level Tiered KV Cache via Mixed-Precision and Sparsification	2	jingwuwa	lixuank		Jingwu Wang	Martin Kang		Zhuoming Chen		
Profiling KV Cache Quantization Overhead in Reasoning LLMs	3	utkrishs	akanshav	rohinida	Utkrish Sharma	Akansha Vijay	Rohini Das	Zhuoming Chen		
Profiling-Guided Optimization of Deep Learning Inference	3	xuhongh	yunzhi12	yunyilin	Xuhong He	Yunzhi Lu	Yunyi Lin	Shushi Hong		
Pruning for Efficient LLM Serving	2	ajseo	cindyliu		AJ Seo	Cindy Liu		Zhuoming Chen		
Reasoning-Aware KV Cache Management for LLM Serving	3	olu	atan3	ziduoy	Owen Lu	Anthony Tan	Ziduo Yi	Shushi Hong		
SAPS: Step-Aware Pruning Schedule for Diffusion LLMs	3	vkshirsa	sairavig	rjohn	Vedanti Kshirsagar	Sai Ravi Teja Ga	Riya Elizabeth Jc	Zhuoming Chen		
Speculative Decoding on Mini-SGLang	3	tianhenl	callumz	hzhang6	Tianheng Lu	Ke (Callum) Zha	Hongyi Zhang	Zepeng Zhao		
Staleness Modeling and Adaptive Sync for Async RL	3	xiaoleg	fengliny	zhuangcg	Xiaole Guo	Fenglin Yu	Zhuangcheng Gu	Shushi Hong		
TA-MPQ: Task-Aware Mixed-Precision Quantization for LLMs with User-Specified Memory Budget	3	hanweny	letianz4	ruiling	Aaron Yu	Letian Zhang	Ruilin Gao	Mingkuan Xu		
Task-Routing Adaptive KV Cache Eviction	3	xueqil	yibok	zhixianz	Xueqi Li	Yibo Kong	Zhixian Zhang	Shushi Hong		
Thought-Aware Hierarchical KV Cache Retrieval for Efficient Reasoning Model Inference	3	skolawol	akouloge	adityag6	Steven Kolawole	Andrew Koulege	Aditya	Haojia Sun		
Tree-Based Speculative Decoding with Dynamic Width Pruning	3	njasuja	neilpuro	acheong	Nishchay Jasuja	Neil Purohit	Alistair Cheong	Mingkuan Xu		
TTIR-GNN: A Graph Neural Network Cost Model for Triton Kernels	1	okathira			Om Kathira			Mengdi Wu		
TurnKV: Turn-Aware Cache Eviction for Multi-Turn LLM Conversations	3	gshah2	pganesh	ypatlola	Gaurav Shah	Pradyut Ganesh	Yagneek Patlola	Zikun Li		
Understanding the Interaction Between KV Cache Eviction and Quantization for Efficient LLM Infer	3	mingtiac	edwardz3	yueyanw	Mingtian Chen	Haohan Zhang	Yueyan Wu	Shushi Hong		
Predictive KV Cache Eviction for Block-Sparse Attention	3	yutarkk	clejang	claireli	Justin Kim	Cody Lejang	Claire Liu	Siyuan Chen		